



Review Article

Artificial intelligence in personalized learning: A global systematic review of current advancements and shaping future opportunities

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ABSTRACT

This study investigates the role of Artificial Intelligence (AI) in personalized learning within tertiary and higher education contexts worldwide. Guided by the PRISMA 2020 framework (Preferred Reporting Items for Systematic Reviews and Meta-Analyses), a systematic literature review of 25 Scopus-indexed articles published between 2019 and 2024 was conducted to map publication trends, identify dominant AI technologies, and synthesize the benefits, challenges, and future directions of AI-enabled personalized learning systems. The findings reveal a rapid shift from early rule-based systems to sophisticated models that integrate machine learning, natural language processing, intelligent tutoring systems, and large language models such as ChatGPT. AI has been shown to enhance student engagement, motivation, and performance by providing adaptive learning pathways, real-time feedback, and tailored content. However, several challenges persist most notably data privacy and ethical concerns, technological infrastructure constraints, educator readiness, and limited scalability across diverse educational contexts. While AI-driven personalization improves learning effectiveness compared to traditional methods, long-term impacts and issues of equity and inclusion remain underexplored. This review highlights the need for ethical frameworks, robust teacher training, and integrating emerging multimodal AI technologies to support more inclusive, sustainable, and human-centered personalized learning ecosystems. The study provides strategic insights for researchers, educators, and policymakers to guide future deployment of AI in education.

1. Introduction

The advancement of educational technology has transformed the paradigms of learning and teaching across various regions of the world (Criollo-C et al., 2024; Fortuna et al., 2024; Samala, Rawas, et al., 2024). Notably, in education, the emergence of practical technologies such as Artificial Intelligence (AI) holds the potential to revolutionize numerous academic domains, especially learning processes, by offering more personalized and adaptive approaches (Handrayani et al., 2023; Nurhikmah et al., 2024; Setyawan et al., 2024). AI provides responsive and

flexible solutions tailored to student needs, enabling personalized learning experiences by identifying learning patterns and adapting to individual learners' unique preferences and comfort levels (Chen et al., 2023; Kim et al., 2022). With its capability to analyze students' learning performance in real-time, AI delivers prompt and precise feedback while identifying areas for improvement (Murtaza et al., 2022).

Personalized learning is a concept that offers tailored learning experiences aligned with the unique profiles of individual students. Integrating AI technologies enables educational systems to respond more effectively to diverse student needs (Castro et al., 2024; Rane, 2024).

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Technologies such as Machine Learning, Natural Language Processing (NLP), Generative AI, and Intelligent Tutoring Systems (ITS) further enhance personalized learning by delivering dynamic and adaptive experiences that meet learners' specific needs (Alqahtani et al., 2023; Lin et al., 2023; H. Yu & Guo, 2023; J. H. Yu & Chauhan, 2024). Machine Learning facilitates the analysis of learning data to understand students' patterns and preferences (Rastrollo-Guerrero et al., 2020), while Natural Language Processing enables more natural interactions between learners and AI-based learning systems, such as chatbots or virtual learning assistants (Gour, 2020).

Historically, personalization in education predates the advent of AI, drawing on pedagogical models such as Montessori's individualized learning (early 1900s), Bloom's mastery learning (Airasian et al., 1971), and Keller's Personalized System of Instruction (Eyre, 2007), which emphasized differentiation, learner autonomy, and formative feedback, yet were difficult to scale in conventional settings. Advancements in computational technologies subsequently enabled digitally supported personalization, starting with computer-assisted instruction (CAI) in the 1980s, learning management systems (LMS) in the 1990s, and rule-based Intelligent Tutoring Systems (ITS) in the 2000s marking the early integration of Artificial Intelligence (AI) into adaptive instruction. This trajectory continued through the 2010s with machine-learning-based data-driven adaptation and has most recently advanced toward large-language-model (LLM) generative AI (e.g., ChatGPT), offering unprecedented conversational support, real-time feedback, and dynamic path recommendations (Barrera Castro et al., 2025; Fan et al., 2023; Romero Alonso et al., 2025; Timimi et al., 2025). Within this ecosystem, generative AI enables the automatic creation of learning materials tailored to students' comprehension levels, ensuring content relevance and progression alignment (Ruiz-Rojas et al., 2023). Complementarily, ITS provide personalized guidance and adapt instructional strategies in real time, much like human tutors, by leveraging learner interaction data (Rizvi, 2023). Through the collaboration of these technologies, AI establishes a learning ecosystem that is interactive and capable of evolving with learners' needs, enhancing motivation, and achieving improved learning outcomes. Through such technologies, AI establishes an interactive learning environment capable of evolving with learners' needs, thereby enhancing motivation and

ultimately improving learning outcomes. Table 4 summarizes this evolution from rule-based tutoring to LLM-enabled conversational agents, while also highlighting emerging directions such as explainable/ethical AI, human-AI orchestration in classrooms, and equity-focused deployment directly informing RQ1–RQ5. An illustrative overview of this AI-driven personalized learning ecosystem is provided in Fig. 1.

The ecosystem depicted in Fig. 1 encompasses the key participants in the landscape of AI-personalized education, a concept adopted from research by (Maghsudi et al., 2021), organized into four main categories: Educators, Learners, Technology Providers, and Policymakers. Each entity plays a vital role in utilizing AI for personalized learning. AI processes data to tailor learning materials, enabling learners to study independently with content that suits their needs, while educators use AI to provide effective instruction. Technology providers ensure accessibility, while policymakers regulate the implementation of AI in education. Overall, AI forms the foundation of a responsive and sustainable learning ecosystem, facilitating adaptive curricula and directed learning management aligned with individual needs.

Although AI technology offers various benefits in creating personalized learning environments, we must address several challenges for optimal implementation. Many current AI-based learning systems are limited to providing recommendations for specific learning materials or activities without fully considering students' cognitive and emotional aspects (Dai & Ke, 2022; Ezzaim et al., 2024; Vistorte et al., 2024). However, understanding students' mental conditions and motivations is crucial for supporting effective and sustainable learning processes.

Furthermore, the integration of AI into conventional education systems faces significant barriers, both in terms of technological infrastructure and curriculum adaptation (Elbanna & Armstrong, 2024). Consequently, aligning new technologies with traditional teaching methods is essential to ensure that AI implementation not only serves as a temporary solution but is also accepted and optimized across various educational institutions (Folkourng & Sakti, 2022; Putra & Arafat, 2024; Putri et al., 2024). On the other hand, excessive reliance on technology may reduce students' ability to think critically and independently, making the role of educators crucial in balancing both classroom and independent learning processes.

Previous research has investigated how AI technologies and

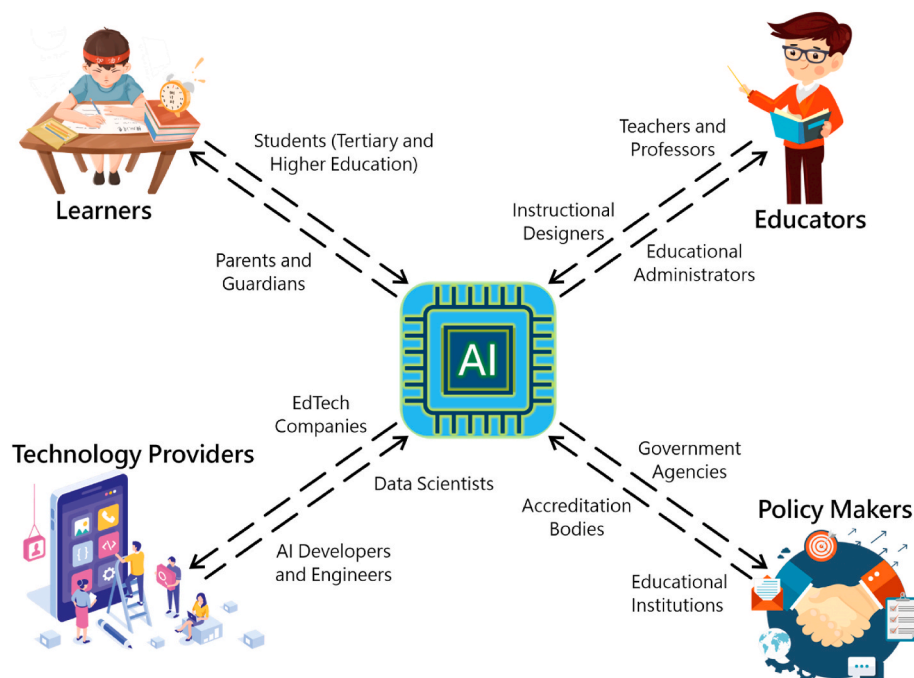


Fig. 1. Baseline ecosystem of artificial intelligence in personalized learning.

techniques are utilized in personalized learning. For example, bibliometric research by (K. C. Li & Wong, 2023) shows a significant growth in AI research trends in personalized learning from 2017 to 2022. Similarly, the presence of Generative AI in research by (Hang et al., 2024) has contributed to automating the creation of multiple-choice questions to support personalized learning. AI technologies in education, such as intelligent tutoring systems and interfaces developed by (Baba et al., 2024), as well as ChatGPT as a conversational AI and natural language processing technology by (González-Rico & Lluch Sintes, 2024), further enrich the learning experience. Therefore, the application of AI in education shows excellent potential to enhance learning effectiveness by providing interactive support responsive to the needs of each individual.

Several prior systematic reviews have explored AI-based personalized learning from different perspectives. For example, (Bhutoria, 2022), mapped the integration of AI and big data in personalized education across the United States, China, and India using a human-in-the-loop model, revealing AI's role in tailoring learning pathways and reshaping teacher roles, while also highlighting challenges related to data privacy and infrastructural inequality. Similarly, (Merino-Campos, 2025), focused on higher education and reported that AI-driven personalized learning can significantly improve academic outcomes through adaptive feedback and data-driven instructional support. Expanding this work into lifelong learning contexts, (Bayly-Castaneda et al., 2024), identified research hotspots in China, India, and the United States, where adaptive technologies and AI-powered path recommendation systems are being leveraged to support skills development in higher education.

More recent reviews have drawn attention to the role of large language models (LLMs) in personalization. Study by (Sharma et al., 2025), emphasize LLMs' potential to enhance student engagement, emotional support, real-time monitoring, and adaptive assessment, while also identifying concerns regarding ethics, bias, privacy, and the importance of authentically simulating pedagogical strategies such as Socratic questioning. At the same time (Barrera Castro et al., 2025), highlight key barriers to inclusive AI deployment, noting that despite rapid advances, the implementation of AI-based personalized learning remains hampered by technological, pedagogical, psychological, institutional, and conceptual challenges which differ across educational levels and modalities. However, none have holistically synthesized empirical evidence on AI-driven personalized learning in the post-ChatGPT era. Given the rapid shift to adaptive systems utilizing deep learning and large-language models since 2019, a focused synthesis of current empirical findings is needed to inform research, practice, and policy. Therefore, this review systematically examines the recent (2019–2024) landscape of AI-based personalized learning within tertiary and higher-education contexts, mapping publication distribution, technological trends, educational impacts, and future research priorities.

Accordingly, this systematic literature review seeks to fill the gap in the literature by offering an up-to-date and holistic overview of trends, benefits, limitations, challenges, implications, and future directions related to the use of AI in personalized learning. Covering studies from the last five years (2019–2024), this timeframe was intentionally selected to ensure the inclusion of the most current developments so that future recommendations can be grounded in relevant empirical evidence. Ultimately, the study aims to promote innovation and effectiveness in implementing personalized learning systems across varied educational contexts. To achieve these objectives, five main research questions are explored:

- RQ1. What is the distribution of research on personalized learning interventions using Artificial Intelligence from 2019 to 2024?
- RQ2. What recent trends have emerged in studies on artificial intelligence (AI) for personalized learning over the past five years?
- RQ3. What are the challenges, opportunities, and benefits of implementing AI-driven personalized learning across various educational contexts?

RQ4. How does AI-based personalized learning impact student performance compared to conventional learning methods?

RQ5. What are the limitations, implications, and future directions identified in studies on AI in personalized learning to improve these systems?

This systematic literature review offers a novel contribution by examining current trends (2019–2024) in AI-based personalized learning an area previously underexplored in existing syntheses, which often predate 2019 or focus on narrow sub-domains. Responding to rapid advancements such as deep learning and large language models that have reshaped personalization design and delivery in education, this study maps recent research and publication distributions while comparing the impact of AI-enabled personalization versus conventional learning methods on student outcomes. It synthesizes the challenges, opportunities, benefits, and risks associated with AI implementation and outlines future directions for optimizing inclusive and effective adaptive learning environments. In addition, it provides strategic recommendations for the development of relevant technologies and education policies. Collectively, these contributions enrich the academic discourse and offer a practical guide for educators, researchers, and policymakers in AI-based education.

2. Methodology

2.1. Research design

This study employs a systematic literature review (SLR) methodology to identify, evaluate, and analyze trends, challenges, opportunities, and future directions of various studies related to the application of artificial intelligence (AI) in personalized learning (Bhatt & Muduli, 2023; Essa et al., 2023; Prasetya et al., 2024). Adopting a PRISMA-aligned approach (Page et al., 2021), the review followed a transparent four-stage process (identification, screening, eligibility, and inclusion), in which Scopus was selected as the primary database due to its multidisciplinary coverage and citation indexing. Boolean keyword searches (“artificial intelligence” AND “personalized learning” AND “education”) were conducted, restricted to peer-reviewed journal articles published in English between 2019 and 2024, and were supplemented by manual backward-reference checking to identify any additional empirical studies.

A qualitative narrative–thematic synthesis was employed to interpret the findings, complemented by bibliometric trend mapping using VOSviewer to visualize the evolution of research topics. Although the PRISMA framework ensures methodological rigor, its predefined inclusion and exclusion criteria (Table 2) may unintentionally omit studies outside traditional quality thresholds (Shamseer et al., 2015). The included articles were examined to explore AI developments over the past five years, with particular attention to challenges, opportunities, and impacts on learner performance. The analysis covered diverse dimensions such as AI methods, implemented technologies, and their effects on engagement, motivation, and learning outcomes in tertiary and higher-education settings (Kuleto et al., 2021; Ouyang et al., 2022). Nonetheless, this focus on established AI approaches may have limited the inclusion of emerging or unconventional techniques that could offer novel insights into personalized learning.

2.2. Data sources and search strategy

The bibliographic search was conducted solely using the Scopus database owing to its comprehensive multidisciplinary journal indexing and metadata filtering capabilities. Initial keyword searches employed combinations of “artificial intelligence,” “personalized learning,” and “education,” with Boolean operators and field-restricted queries (TITLE-ABS-KEY). Filters were applied for year range (2019–2024), document type (journal articles), peer-review status (final stage publications), and

language (English). Although publisher platforms (e.g., Emerald Insight, IEEE Xplore, MDPI, ScienceDirect, SpringerLink, Wiley Online Library) were consulted only to access full texts of Scopus-indexed articles, they were not used as independent databases for article retrieval. The detailed search query and retrieval date (31 December 2024) are reported in Table 1.

Table 1 presents the data sources and keywords utilized in the literature search for this study. In the initial phase, keyword searches were conducted using terms such as “artificial intelligence,” “personalized learning,” and “education,” with the search process carried out systematically using filters based on a publication time frame of the last five years, from 2019 to 2024.

To ensure historical continuity and frontier relevance, we incorporated both foundational studies on early intelligent tutoring and machine-learning-based personalization (Lalit et al., 2025; Zhong et al., 2024) and the most recent advances in large-language-model-based generative AI, explainable AI, and equity-aware educational analytics (Rodríguez-Ortiz et al., 2025). This broader reference integration enables a more comprehensive synthesis of past progress and emerging trajectories in AI-driven personalized learning.

2.3. Inclusion and exclusion criteria

This systematic literature review exclusively retrieved records via the Scopus database search engine, which aggregates journal content from multiple indexed publishers (e.g., Emerald, IEEE, MDPI, SpringerLink, Wiley). To complement this, manual backward reference-list checks were conducted to identify additional potentially relevant studies. We applied strict inclusion and exclusion criteria to ensure methodological rigor and replicability, as outlined in Table 2.

Only original empirical research articles (quantitative, qualitative, or mixed-methods) published in peer-reviewed journals, written in English, and finalized for publication between 2019 and 2024 were included. Articles had to explicitly mention Artificial Intelligence and Personalized Learning in the title, abstract, or keywords, and focus on educational contexts. Conversely, we excluded editorials, conference papers, book chapters, preprints, conceptual or theoretical studies, as well as articles focusing on personalized learning without a clear AI component.

After applying these criteria, the initial Scopus search results were identified through manual reference-list searches, duplicate removal, and screening processes, all guided by the PRISMA 2020 flow diagram (Fig. 2).

Fig. 2 illustrates the article selection process following the PRISMA flow diagram. In the identification phase, 1922 records were identified from Scopus, consisting of 1661 from the database and 261 from the register. After removing 49 duplicate records, 954 articles were

Table 1
Data source information (Systematic Literature Review).

Data Source	Description
Database	Scopus
Initial keyword search	(TITLE-ABS-KEY (artificial AND intelligence) AND TITLE-ABS-KEY (personalized AND learning) AND TITLE-ABS-KEY (education)) AND PUBYEAR >2014 AND PUBYEAR <2024
Keyword search (with restrictions)	(TITLE-ABS-KEY (artificial AND intelligence) AND TITLE-ABS-KEY (personalized AND learning) AND TITLE-ABS-KEY (education)) AND PUBYEAR >2019 AND PUBYEAR <2024 AND (LIMIT-TO (DOCTYPE, "ar")) AND (LIMIT-TO (EXACT KEYWORD, "Artificial Intelligence")) OR LIMIT-TO (EXACT KEYWORD, "Personalized Learning")) AND (LIMIT-TO (LANGUAGE, "English")) AND (LIMIT-TO (SRCTYPE, "j")) AND (LIMIT-TO (PUBSTAGE, "final"))
Timeframe and Language	2019–2024 and English article

Table 2
Inclusion and exclusion criteria for screening articles.

Criteria	Inclusion (Accepted Articles)	Exclusion (Rejected Articles)
Publication Type and Stage	Original research articles published in scientific journals have gone through the final publication stage.	Editorials, conference proceedings, book series, trade journals, and articles are still in press.
Language	Articles published in English.	Articles published in languages other than English.
Time Frame	Articles published between 2019 and 2024.	Articles published before 2019.
Keywords and Scope	Articles specifically mentioning Artificial Intelligence and Personalized Learning in the title, abstract, or keywords.	Studies focusing on personalized learning without an emphasis on Artificial Intelligence.
Research Method	Empirical studies (quantitative, qualitative, or mixed-methods).	Conceptual papers or articles without empirical data, such as review articles.
Relevance of Research Focus	Articles explicitly discussing the application of Artificial Intelligence in Personalized Learning in education.	Research not directly related to education or AI applications outside the context of personalized learning.
Data Quality	Clearly defined objectives, detailed methodology, and measurable outcomes.	Unclear research design, low data quality, or insufficient reporting.

automatically excluded by the screening tools, and 268 were removed for other reasons, such as conference papers or articles still in press. This left 661 records for the screening phase.

In the screening phase, 134 records were excluded by automation tools used in the Scopus screening, and 254 reports were not retrieved due to access issues, leaving 407 reports for further evaluation. Following the eligibility assessment, 135 reports remained, but 34 were excluded for being conceptual papers or reviews, 27 for not being related to education or AI in personalized learning, and 49 for lacking sufficient data or having unclear results. Ultimately, 25 studies met all the inclusion criteria and were included in the final review. This process shows how the criteria in Table 2 were applied, ensuring that only the most relevant and high-quality studies were included in the final review.

2.4. Quality assurance process

To ensure this systematic review’s quality, transparency, and reliability, we implemented a structured Quality Assurance (QA) protocol across all stages. Initially, a pilot round was conducted to calibrate the screening form and codebook. Two reviewers independently performed title/abstract and full-text screening according to predefined inclusion/exclusion criteria (Table 2), following PRISMA guidelines (Fig. 2). Reasons for exclusion were documented, and any discrepancies were resolved through discussion or with a third reviewer if needed.

For data extraction, two reviewers independently extracted study characteristics and outcomes using a standardized coding template aligned with the research questions (RQ1–RQ5). Extracted entries were cross-checked, reconciled by consensus, and assessed for methodological rigor using the embedded “Data Quality” rubric. Only studies meeting these standards were included in the final dataset, which was jointly validated before analysis.

Data synthesis combined narrative/thematic approaches with bibliometric visualizations using VOSviewer and Microsoft Excel. The research team documented parameter settings (e.g., keyword normalization, clustering options) to ensure reproducibility. All stages (screening, extraction, synthesis) were performed collaboratively to minimize bias and ensure consistency, with dual independent review and arbitration as necessary. The complete search strings, coding templates, and decision logs are available from the corresponding author upon reasonable request to enhance transparency and replicability.

To further strengthen transparency in line with open science

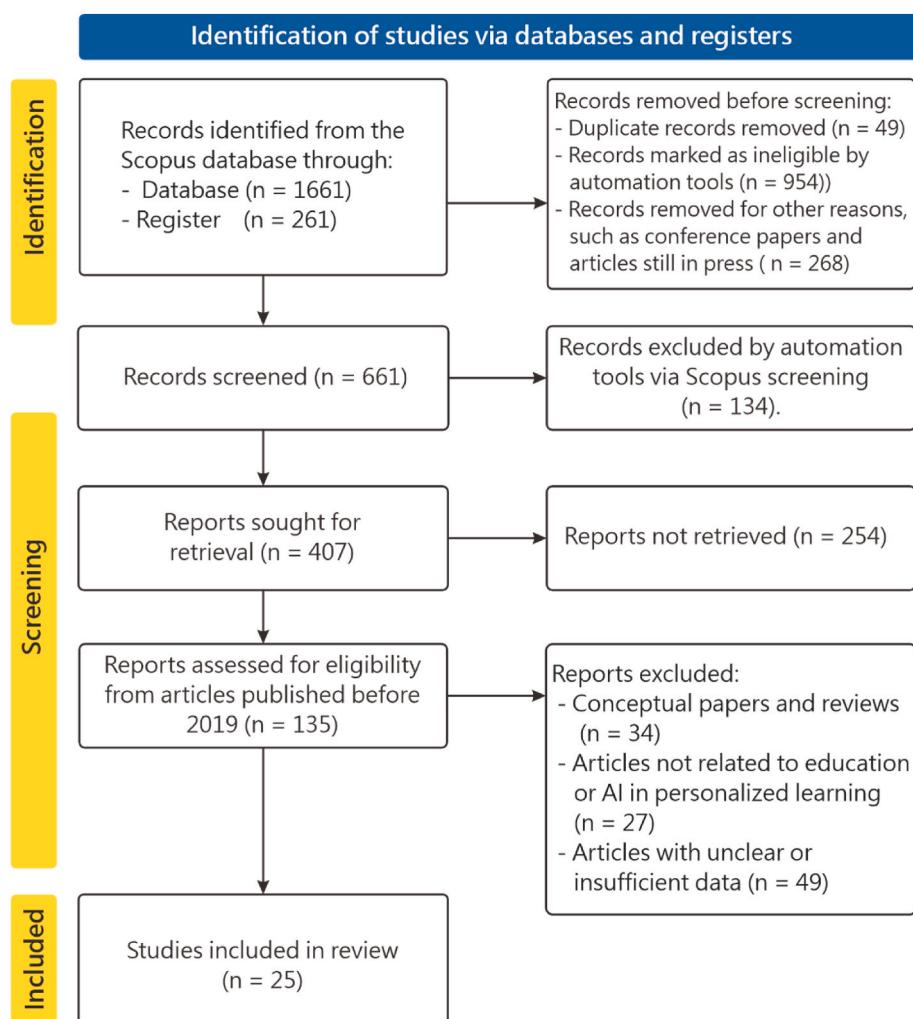


Fig. 2. PRISMA flow diagram for artificial intelligence in personalized learning.

practices, all coding sheets and extraction files are available from the corresponding author upon reasonable request. This rigorous QA process helped ensure the robustness, reproducibility, and trustworthiness of the review findings.

2.5. Data analysis technique

The data analysis in this study systematically synthesizes findings from diverse studies on Artificial Intelligence (AI) in personalized learning. Extracted results were processed using Microsoft Excel and summarized through graphs, flow diagrams, and tables to enhance readers' understanding of prevailing trends, opportunities, and challenges (Prasetyo et al., 2024; Rozan et al., 2024; Samala, Bojic, et al., 2024). These outputs were deliberately presented in transparent visual formats to clearly convey the evolution of AI-based personalized learning.

In addition, bibliometric techniques were employed using VOSviewer to explore keyword co-occurrence networks from RIS-formatted datasets. This visualization approach facilitated the identification of emerging topics, clusters, and relationships among variables such as technology, pedagogy, and learning outcomes, which are central to AI development in education (Bukar et al., 2023; McAllister et al., 2022). Clustering patterns generated through VOSviewer (e.g., based on keyword normalization and Leiden algorithms) highlighted research groupings that helped reveal gaps and inform suggestions for future investigation (Fortuna et al., 2024; Muskhair et al., 2023; Prasetya et al.,

2025).

This review employed a primarily qualitative narrative and bibliometric approach. Given the considerable heterogeneity in study designs, intervention types, and reported outcomes across the included articles, conducting a formal statistical meta-analysis or regression-based synthesis was not deemed appropriate. Instead, descriptive statistics (e.g., counts, frequencies, and percentages) were calculated via Microsoft Excel to summarize topical distributions, while narrative thematic analysis was used to synthesize qualitative findings. Bibliometric visualizations were produced using VOSviewer with standard co-occurrence thresholds (minimum keyword occurrences = 2). No inferential statistical testing (such as p-values or confidence intervals) was carried out.

3. Results

3.1. Distribution of research on personalized learning interventions using artificial intelligence (2019–2024)

The distribution of research on personalized learning interventions using AI demonstrates a significant increase from 2019 to 2024, based on data from international journals indexed in the Scopus database, as shown in Fig. 3. This study reveals various AI technologies, such as Deep Learning, Machine Learning, Natural Language Processing (NLP), and Intelligent Tutoring Systems. These technologies enable AI to tailor instructional content to students' pace and learning style and provide real-time feedback, accelerating the learning process. Since the advent

of ChatGPT in 2022, this technology has garnered attention due to its ability to process large-scale data and deliver more relevant and personalized responses (Nazir & Wang, 2023). The distribution of publications related to AI in personalized learning is visualized in Fig. 3 in response to Research Question 1 (RQ1).

Fig. 3 presents an infographic showing a significant increase in research on personalized learning using AI over the past five years. The most significant surge was observed during 2023–2024, when the research distribution doubled. This emphasizes the high level of attention and relevance of this topic among researchers, educators, and policymakers seeking to integrate AI technology into various aspects of education.

The data in the infographic also shows a positive trend, with the field of Computer Science dominating the research contribution at 24 %. However, other fields with contributions under 20 % indicate significant growth potential, such as social sciences (20 %), engineering (12 %), medicine (10 %), and other multidisciplinary applications (20 %). This reflects that the application of AI in personalized learning is not only relevant to information technology but is also expanding into various disciplines requiring adaptive learning solutions.

For instance, in the social sciences, AI can be used to analyze student behavior, while in engineering, this technology supports the creation of project-based learning simulations. In medicine, AI assists in personalizing the curriculum based on students' understanding levels and practical needs. By combining cross-disciplinary expertise, future research has the potential to create a more inclusive, adaptive, and responsive learning ecosystem that meets individual needs. Therefore, it is crucial to continue encouraging exploration in underrepresented fields to realize the full potential of AI in personalized learning.

From an initial pool of 1922 records, only 25 empirical studies met all predefined inclusion criteria following duplicate removal, relevance screening, and methodological appraisal. This rigorous selection process ensured that the final review focused solely on high-quality, peer-reviewed articles, thereby enhancing the reliability of the synthesized insights into current trends, geographic distribution, and research impact in AI-based personalized learning. For transparency, citation counts are presented using two metrics: (i) Scopus citations to reflect impact within the indexed scholarly literature, and (ii) global citations retrieved from Google Scholar, which captures wider influence across conference papers, theses, and other grey literature. Details of the included studies, including domain, source journal, and citation indicators, are summarized in Table 3.

Table 3 provides an overview of AI-based personalized learning across various research domains, highlighting studies from countries such as Taiwan, Kazakhstan, the United States, and China, which focus on the application of AI in personalized learning environments. For instance (Huang et al., 2023), from Taiwan explored AI-driven personalized video recommendations, garnered 314 citations in Scopus, and

522 global citations. Similarly (Tapalova & Zhiyenbayeva, 2022), from Kazakhstan focus on personalized learning pathways, with 280 citations in Scopus and 727 global citations. The table emphasizes the diversity of AI applications, ranging from personalized feedback systems (Xu et al., 2023) to AI in fitness training (Mokmin, 2020), and identifies citation metrics as indicators of the impact and recognition of each research effort. Some newer works, such as those by (Guettala et al., 2024; Hang et al., 2024), reflect emerging research on the potential of AI to enhance adaptive learning environments despite having fewer citations. Consequently, the findings answer (RQ1), showing that the distribution of research has experienced significant growth in both volume and scope, reflecting a global trend in exploring AI's potential to create more personalized and adaptive learning environments.

3.2. Emerging trends in AI for personalized learning (2019–2024)

The rapid development of artificial intelligence (AI) technologies in education has brought about significant growth and innovation in personalized learning systems that respond directly to individual student needs. The current study visualizes these emerging trends using VOSviewer, illustrating how AI is increasingly being applied to deliver adaptive learning pathways, intelligent content recommendations, and real-time feedback in line with RQ2. To better understand these developments, it is essential to examine how the field has evolved chronologically. The historical progression of AI-driven personalized learning demonstrates a clear transition from early approaches focused on simple rule-based personalization towards more complex models utilizing deep learning, real-time analytics, and most recently, generative AI. Table 4 provides a timeline of key milestones and representative studies from 2019 to 2024, highlighting how these technological advancements have progressively expanded the capabilities and applications of personalized learning systems.

Table 4 demonstrates a progressive shift in AI-driven personalized learning from rule-based, learning-style personalization (2019) toward increasingly sophisticated adaptive systems. In 2020, the integration of AI into virtualized and physical learning environments marked the introduction of adaptive and context-aware systems. By 2021, the field began leveraging big data infrastructures, cloud computing, and intelligent algorithms such as LinUCB and artificial neural networks) to enhance precision and scalability. The year 2022 signaled the adoption of deep learning frameworks in personalized learning pathways, further strengthening the system's capacity to tailor instruction dynamically. In 2023, research attention moved toward real-time personalized feedback, natural language processing, and predictive analytics, reflecting the growing use of intelligent technologies to support immediate and adaptive decision-making in learning environments. Most recently, in 2024, generative AI and large language models such as ChatGPT triggered a new phase focused on autonomous content generation, smart

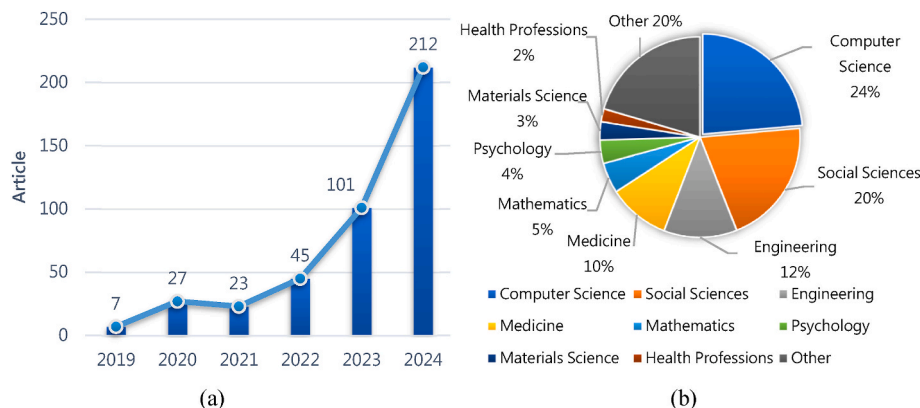


Fig. 3. a) Annual growth in AI-personalized learning publications (2019–2024); b) Distribution of research topics across domains.

Table 3
Summary of included studies (Domain, source, citation impact).

No	Author/Country	Research Domain	Journal Source	Citations by Scopus	Global Citations
1	(Huang et al., 2023)/(Taiwan)	AI-enabled personalized video recommendations	Computers and Education (Elsevier)	314	522
2	(Tapalova & Zhiyenbayeva, 2022)/(Kazakhstan)	AIEd for Personalized Learning Pathways	The Electronic Journal of e-Learning	280	727
3	(Tang et al., 2020)/(United States)	Personalized learning system for parallel intelligent education	IEEE Transactions on Computational Social Systems	83	116
4	(Xiao & Yi, 2021)/(China)	Artificial Intelligence for personalize education reform	Computer Applications in Engineering Education (Wiley Online Library)	57	107
5	(Alsobhi & Alyoubi, 2019)/(Saudi Arabia)	AI is learning to personalize education based on learning styles and dyslexia types.	Data Technologies and Applications (Emerald Insight)	30	53
6	(Xu et al., 2023)/(China)	AI-assisted personalized feedback system	International Journal of Modeling, Simulation, and Scientific Computing	37	56
7	(Mokmin, 2020)/(Malaysia)	AI algorithms for personalized recommendations for a virtual fitness trainer	International Journal of Human Movement and Sports Sciences	31	50
8	(Wei et al., 2021)/(China)	LinUCB-based AI is proposed in an algorithm that explores personalized coefficients.	Frontiers in Psychology	32	64
9	(Bekmanova et al., 2021)/(Kazakhstan)	AI in distance learning platform enhancement involving artificial neural networks	Journal of Computing in Higher Education (Springer Nature)	21	52
10	(Sajja et al., 2024)/(United States)	AI-enabled personalized learning experiences involving natural language processing techniques	Information (MDPI)	122	341
11	(Zhang, 2022)/(China)	AI in personalized learning platforms involves technology like deep learning, neural networks, and TensorFlow	Wireless Communications and Mobile Computing (Hindawi)	18	25
12	(Ogata et al., 2024)/(Japan)	AI-supported e-learning systems for personalized learning	Research and Practice in Technology-Enhanced Learning	31	50
13	(Wang, 2023)/(China)	AI-based personalized learning resource recommendation algorithm	International Journal of Computer Applications in Technology (Inderscience)	18	19
14	(Ruwe & Mayweg-Paus, 2023)/(Germany)	AI in educational environments in processing personalized natural language	Computers and Education: Artificial Intelligence (Elsevier)	14	34
15	(Kucirkova & Leaton Gray, 2023)/(Norwegia)	AI learning encompasses interconnected technologies, including deep learning, machine learning, neural networks, and predictive analysis.	Educational Theory (Wiley Online Library)	11	26
16	(Baba et al., 2024)/(Morocco)	AI-driven personalized learning environments employing intelligent tutoring systems and chat interfaces	International Journal of Interactive Mobile Technologies (IJIM)	21	38
17	(Naseer et al., 2024)/(Pakistan)	AI in personalized education is mainly deep learning	Heliyon (Elsevier)	41	77
18	(Kanchon et al., 2024)/(Bangladesh)	AI-driven learning style identification in personalized learning	International Journal of Cognitive Computing in Engineering (Elsevier)	22	58
19	(González-Rico & Lluch Sintes, 2024)/(Spain)	ChatGPT in conversational AI and natural language processing technologies	Education Sciences (MDPI)	6	16
20	(Baltezarević & Baltezarević, 2024)/(Serbia)	AI in personalized education primarily involves machine learning and deep learning	International Journal of Cognitive Research in Science, Engineering and Education	7	17
21	(Hang et al., 2024)/(Hong Kong)	Generative AI in automating multiple-choice question generation for personalized learning	IEEE Access	41	48
22	(Guettala et al., 2024)/(Algeria)	The transformative potential of generative AI in creating adaptive and personalized learning environments	Acta Informatica Pragensia	–	64
23	(Dong et al., 2024)/(China)	AI in IoT integration and Generative AI for English education	Journal of Computational Science (Elsevier)	13	35
24	(Annus & Kmeť, 2024)/(Slovakia)	Strengthening AI personalized learning in mathematics education	Education Sciences (MDPI)	3	9
25	(Deng et al., 2024)/(China)	AI in Optimizing Personalized Learning Pathways	International Journal of Advanced Computer Science and Applications	11	17

tutoring, and pathway optimization through technologies like ITS, IoT integration, and conversational AI.

To strengthen this understanding of how research focus has evolved over time, it is valuable to examine not only the chronological progression of technologies, but also the shifting prominence of thematic concepts within the literature. The keyword network visualizations and their evolution over the past five years are presented in Figs. 4 and 5.

Based on the VOSviewer visualization in Fig. 4, recent developments in the application of Artificial Intelligence (AI) for personalized learning reveal several significant trends, showing strong interconnections between key topics such as artificial intelligence, personalized learning, machine learning, students, and education. The red cluster highlights AI's central role as the network's core, surrounded by terms like machine learning, education, and ChatGPT. These findings reflect a strong focus on the development of AI-based algorithms and tools to support education. Nevertheless, a research gap exists in integrating AI with cross-disciplinary contexts such as psychology and pedagogy. Future

studies could explore how AI can model cognitive and emotional aspects to enhance students' learning experiences.

The green cluster includes terms such as personalized learning, students, e-learning, and educational technology, indicating an interest in using adaptive technologies to personalize the learning experience. Nevertheless, the long-term impact of personalized learning on educational equity and inclusion has received limited attention. Therefore, future research may explore the potential of AI to support underserved communities and reduce disparities in educational access.

The blue cluster comprises terms like recommendation systems, privacy, and ethics, reflecting concerns regarding the responsible application of AI in education. However, ethical issues like algorithmic bias and data privacy protection have not been explored as extensively as technical innovations. Future research could expand the development of ethical frameworks to ensure fair and transparent AI applications in education.

The yellow cluster highlights teaching, curriculum, and teacher

Table 4
Timeline of historical developments in AI-driven personalized learning (2019–2024).

Year	Key Developments & Representative Studies
2019	Initial focus on personalization based on learning styles and dyslexia types (Alsobhi & Alyoubi, 2019).
2020	Expansion toward adaptive systems and AI-based physical training: - (Tang et al., 2020) – Parallel Intelligent Education. - (Mokmin, 2020) – Virtual Fitness Trainer for personalized physical activity.
2021	Focus on integrating AI + Big Data + Cloud and increasing algorithm accuracy: - (Xiao & Yi, 2021) – AI educational reform. - (Bekmanova et al., 2021) – blended personalized learning via ANNs. - (Wei et al., 2021) – LinUCB algorithm for personalized recommendations.
2022	Adoption of deep learning frameworks in modern learning platforms: - (Tapalova & Zhiyenbayeva, 2022) – personalized pathways. - (Zhang, 2022) – TensorFlow/Deep Learning for personalized learning.
2023	Era of real-time feedback, NLP and predictive analytics: - (Huang et al., 2023) – AI video recommendation in flipped classroom. - (Xu et al., 2023) – personalized feedback system (PFS). - (Kucirkova & Leaton Gray, 2023 ; Ruwe & Mayweg-Paus, 2023 ; Wang, 2023) – NLP, resource recommendation, predictive analytics.
2024	Boom of generative AI and LLMs (ChatGPT) in adaptive learning: - (Guettala et al., 2024 ; Hang et al., 2024 ; Sajja et al., 2024) – Generative AI for adaptive MCQs and learning environments. - (González-Rico & Lluch Sintes, 2024) – use of ChatGPT for soft skills development. - (Dong et al., 2024) – integration of AI + IoT. - (Annuš & Kmeť, 2024 ; Baba et al., 2024 ; Deng et al., 2024 ; Ogata et al., 2024) – ITS, e-learning personalization, pathway optimization.

topics, focusing on the interaction between educators and AI-based systems. While there is interest in the potential of AI to support educators, research on educators’ perceptions of this technology and their readiness to integrate it into teaching practices remains limited. Future studies could focus on developing training programs for educators to enhance human-machine collaboration in classroom learning.

Overall, this visualization provides insights into the dynamics of AI research development in personalized learning. While there are several gaps, such as cross-disciplinary integration, scalability, implementation, long-term studies, and sustainability, future research should aim at more inclusive, sustainable, and broadly impactful solutions. This approach

will optimize AI’s transformative educational potential to create more adaptive and responsive learning systems globally.

Fig. 5 presents a bibliometric visualization generated using VOSviewer, illustrating the current research trends related to the application of Artificial Intelligence (AI) in personalized learning. This visualization depicts the relationships between key terms and the distribution of research from 2019 to 2024. Yellow indicates increasingly popular themes in recent years, such as ChatGPT and contrastive learning, while green highlights topics developed earlier but remain relevant, such as machine learning and personalized learning.

The term “artificial intelligence” occupies a central position in the network, with extensive connections to various other terms, reflecting AI’s role as the core of this research. Furthermore, its linkage to terms like ethics, privacy, e-learning, and students underscores attention to multidisciplinary challenges, including ethical concerns regarding AI use, data privacy protection, and the integration of educational technologies to support more personalized and adaptive learning.

Although research has made significant progress, several gaps remain to be addressed. For instance, privacy and data security issues have not received adequate attention. Similarly, accessibility and inclusion for underserved communities remain significant challenges. Moreover, the long-term impact of AI on academic performance and students’ well-being has yet to be explored in depth. The relationship between AI and the teacher’s role in curriculum design and creating more adaptive teaching methods also requires further attention in future studies.

Prospects in this field include the development of ethical frameworks that ensure the responsible use of AI, integrating AI with local needs to expand educational access, and using AI-based technologies, such as ChatGPT, to enhance content personalization and teaching methods. Research on these dimensions is expected to yield innovative, inclusive, and sustainable solutions that can transform the global education ecosystem. Thus, these findings answer (RQ2) by demonstrating that over the past five years, AI has served as a supporting technology and has become a key driver in creating student-centered, equitable, and globally responsive learning systems.

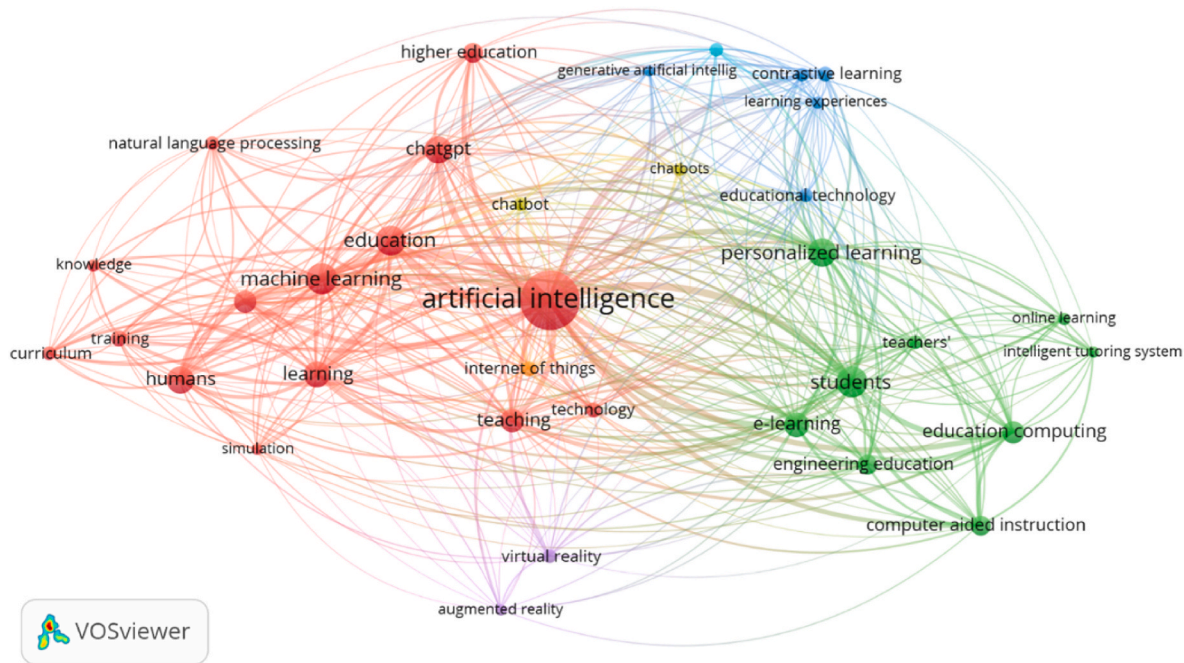


Fig. 4. Visualization of research topic connections in AI for personalized learning.

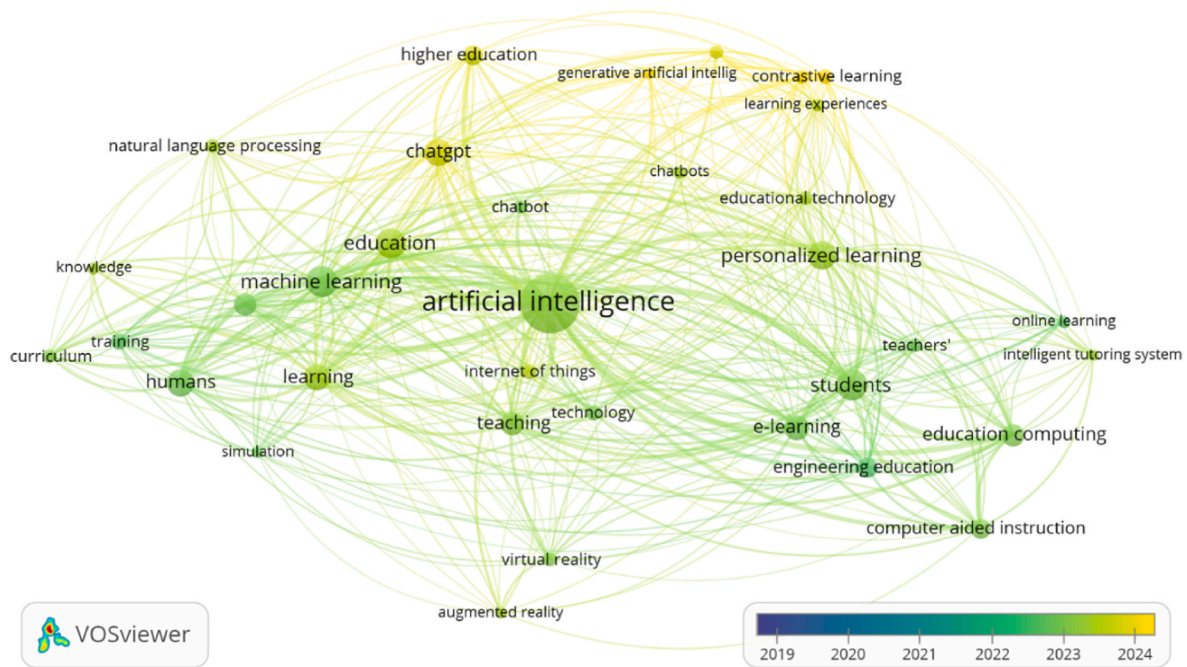


Fig. 5. Trends and research focus on AI for personalized learning (2019–2024).

3.3. Challenges, opportunities, and benefits of AI-driven personalized learning

In order to address the research question (RQ3) concerning the challenges, opportunities, and benefits of Artificial Intelligence-driven personalized learning, Table 5 summarizes key findings from recent research across various educational contexts.

Table 5 provides a comprehensive overview of the challenges, opportunities, and benefits of implementing AI-based personalized learning across various educational institutions, including the need for adequate technological infrastructure, educator training, and data privacy and AI bias issues. Furthermore, the opportunities arising from AI implementation include the personalization of educational content to meet individual student needs, improved accessibility, and the use of big data analytics to enhance educational resource management.

The benefits include increased student motivation following AI technology, improved learning outcomes through real-time feedback, and a more adaptive and personalized learning experience. In addressing the research question (RQ3), Table 5 successfully presents concrete evidence from a range of studies demonstrating that, despite technical and ethical challenges, AI can enhance educational quality through more effective personalization and increased student engagement in the learning process.

3.4. Impact of AI-based personalized learning on student performance

AI-based learning has proven more effective in improving student performance than traditional methods, which often provide uniform content for all students with slow feedback. This results in students with slower learning speeds falling behind, while students with higher capabilities are not adequately challenged to develop their potential. In contrast, AI-based learning can tailor the learning experience to the needs and abilities of each student, as shown in Fig. 6.

As depicted in Fig. 6, the AI-based framework facilitates a more individualized and measurable approach. AI analyzes student learning patterns in real-time, provides instant feedback, and adjusts content according to each individual's understanding level and learning style. The four key elements of the AI framework for personalized learning are:

- Customized learning paths involve AI automatically adjusting the learning trajectory based on a student's mastery of a specific topic. For example, several students struggling with basic mathematical concepts will be provided additional material until sufficient understanding is reached before progressing to the next level.
- Competency-based learning allows students to proceed to the following material after fully mastering the previous content. Unlike conventional methods, this ensures a deep understanding of concepts, where student progress is often determined by a rigid schedule rather than individual comprehension levels.
- AI-based Systems that Maintain Dynamic Student and Teacher Profiles track each student's learning preferences, strengths, and weaknesses. This data enables the alignment of teaching methods with the student's learning style. For instance, visual learners will receive more graphics and diagrams, while verbal learners will benefit from text-based explanations.
- Personalized feedback mechanisms enable learners to identify and correct their mistakes quickly, thanks to AI technology. Unlike traditional methods, where teacher feedback is often delayed until long after completing assignments, AI provides immediate correction and guidance.

Recent studies (Demartini et al., 2024; Gligorea et al., 2023) indicate that implementing machine learning in adaptive learning systems has improved student understanding by up to 25 % compared to conventional learning methods. For example, adaptive learning algorithms used in mathematics lessons adjust the difficulty level of questions based on student responses, so more proficient students are given more complex challenges, while those needing additional support receive more straightforward questions with supplementary explanations. The core AI technologies in personalized learning include:

- Machine Learning Models analyze patterns in student performance and learning behaviors, enabling predictions about areas where students need additional support or learning resources.
- Natural Language Processing (NLP) allows the AI system to understand and respond to student queries in natural language, creating a more interactive learning experience by providing instant feedback and accelerating material comprehension.

Table 5
Challenges, opportunities, and benefits of AI personalized learning systems.

No	Author	Challenges	Opportunities	Benefits
1	Huang et al. (2023)	There was no significant difference in learning motivation between the control and experimental groups.	Opportunity to increase learning motivation through an AI-based recommendation system in flipped classrooms.	Increased learning motivation in the experimental group, supported by independent iLearning.
2	Tapalova and Zhiyenbayeva (2022)	There is a need for technological infrastructure and educator training to integrate AIED (Artificial Intelligence in Education).	Personalization and adaptation of educational content according to student needs and increased access to education.	Improved engagement, instant data analysis, and development of 21st-century competencies.
3	Tang et al. (2020)	Difficulty in providing resources to support personalized learning frameworks.	Parallel innovative education allows virtual learning to support personalization and differentiation.	Enhance strategic thinking and problem-solving skills in students.
4	Xiao and Yi (2021)	Challenges in integrating AI, big data, cloud computing, and IoT in personalized learning.	Opportunity to design personalized training plans applicable to careers or other areas.	AI-based educational reform to improve higher education quality.
5	Alsobhi and Alyoubi (2019)	Experimental results are limited to computer science students and have not been tested in a broader educational context.	Development of adaptive systems with more personalization variables and validation in various contexts.	Improved learning outcomes through personalization based on learning style and dyslexia type.
6	Xu et al. (2023)	Ethical and accurate collection and analysis of student data	AI-PFS can enhance academic interaction and collaboration and provide timely, relevant feedback.	Improved student performance by 95.32 %, highlighting the effectiveness of personalized feedback in enhancing educational outcomes.
7	Mokmin (2020)	Limited testing on specific user groups, requiring broader trials and reduced prediction errors through more data	Development of personalized, accurate virtual trainers that boost user engagement in physical activities.	Increased physical activity engagement without teacher presence, with potential for designing practical and appealing virtual fitness trainers.
8	Wei et al. (2021)	Designing recommendation schemes that consider individual learning capabilities	Use AI and educational psychology to personalize learning resources, enhancing student potential.	Improved accuracy, adaptability, and personalization in resource recommendations, reducing cognitive load and matching learning materials with student abilities.
9	Bekmanova et al. (2021)	Adapting learning models to individual student needs, especially in distance and blended learning contexts.	Use of personalized training models leveraging data science methods like ontology modeling.	Enhanced course effectiveness and flexibility, as shown in "Digital Skills of a Modern Teacher in Distance Learning" evaluations.
10	Sajja et al. (2024)	Difficulty measuring accuracy and perceived benefits without a comprehensive evaluation	Potential to enhance natural language understanding and learning personalization through adaptive algorithms.	Improved student outcomes, engagement, and satisfaction, contributing to broader discussions on AI and NLP integration in education.
11	Zhang (2022)	An adequate technological infrastructure and educator training for AI-based platforms are needed.	AI can collect large datasets and provide actionable insights, improving teaching methods and tailoring materials to student needs.	Enhanced learning efficiency, personalized education services, and better outcomes, boosting creativity for students and teachers.
12	Ogata et al. (2024)	Integrating explainable AI into personalized learning systems while ensuring comprehensible explanations	Automatically predict and group student personalities based on learning logs, improving recommendation effectiveness.	Personalized quiz recommendations and tailored explanations aligned with student concept mastery, enhancing learning experiences and outcomes.
13	Wang (2023)	Integration of IoT in education systems and the need for teacher/student training	Significant potential in big data collection and analysis for improving learning and resource management	Increased efficiency in learning resource recommendations
14	Ruwe and Mayweg-Paus (2023)	Assessing motivation in short-duration interventions and generalization of results	Exploring AI feedback across different contexts and its impact on learner reactions	AI systems can be more trusted than peers or teachers, improving feedback.
15	Kucirkova and Leaton Gray (2023)	Balancing AI with human intelligence in personalized learning	Innovative AI education systems can enhance civic participation and human well-being	AI can improve the monitoring of student performance and learning outcomes
16	Baba et al. (2024)	Potential bias and inequality in AI-driven personalized learning	Significant improvements in student engagement, achievement, and knowledge retention	Enhanced student experiences, motivation, and academic outcomes
17	Naseer et al. (2024)	Technical integration issues and resistance to new technology	Improved academic performance and student engagement, offering a practical framework for adaptive teaching	Contributing to increased demand for AI in education and valuable insights for developers
18	Kanchon et al. (2024)	Need for multimodal data integration for better personalized learning	Opportunities for integrating learning style detection and content modification with standard learning management systems	Enhanced personalized learning experiences and academic performance
19	González-Rico and Lluch Sintés (2024)	Need for larger, diverse samples for statistical analysis	Integration of AI-based digital tools with personal tutor interaction to improve soft skills development	Increased relevance of academic content, better preparing students for the challenges of a digital and global world
20	Baltezarević and Baltezarević (2024)	Concerns about data privacy, AI bias, and reduced educator roles in personalized learning	AI's potential to personalize learning experiences, improve academic achievement	Enhanced accessibility, efficiency, and innovative teaching resources, with AI identifying areas of student difficulty
21	Hang et al. (2024)	Difficulty in creating effective MCQs	Leveraging large language models (LLMs) to generate more targeted, educationally relevant MCQs	Personalized learning experiences and improved student performance, with more challenging and relevant MCQs
22	Guettala et al. (2024)	Ethical issues and reduced human interaction in learning	Generative AI offers significant opportunities for personalized, adaptive learning experiences	Increased student engagement, reduced educator workload, and scalable real-time learning adjustments
23	Dong et al. (2024)	Dependence on quality data and significant computational resources for AI applications	Integration of IoT and generative AI in English education to create adaptive, culturally sensitive learning	Enhanced learning outcomes with diverse, tailored English learning content and accurate speaking feedback

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Table 5 (continued)

No	Author	Challenges	Opportunities	Benefits
24	Annus and Kmeř (2024)	Small sample sizes and lack of control groups affect external validity	Opportunities for developing more effective educational software, especially for younger age groups	Improved math motivation and understanding, as shown by better test scores and positive feedback from teachers
25	Deng et al. (2024)	Increased time costs are affecting the feasibility of AI algorithms in resource-limited environments	Introduction of dynamic models and incremental learning algorithms for more effective personalized learning paths	Improved learning effectiveness, offering more personalized and equitable educational services

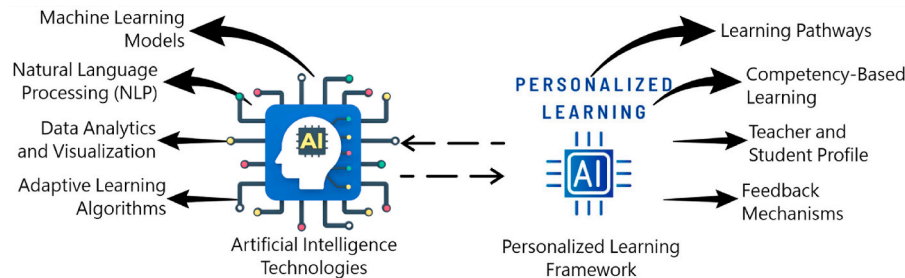


Fig. 6. Core components of AI technologies and a personalized learning framework.

- c. Data analysis and visualization involve presenting student progress in clear and accessible formats, such as charts or interactive maps. These tools assist educators in making precise decisions by providing actionable insights. For example, they enable teachers to identify students who require special attention, allowing for the effective implementation of targeted teaching strategies to address individual learning needs.
- d. Adaptive Learning Algorithms adjust the difficulty of materials based on the student’s understanding and development level.

In summary, AI technology and the personalized learning framework offer innovative approaches to addressing various challenges in traditional learning, particularly in responding to research questions (RQ4). Leveraging cutting-edge technologies such as machine learning models, natural language processing (NLP), and adaptive learning algorithms, this framework can deliver a more efficient, relevant, and focused learning experience. Compared to conventional methods, AI-based learning deepens students’ understanding of concepts and fosters active engagement in the learning process. This advantage highlights the potential of AI in creating a more effective and impactful educational ecosystem.

3.5. Limitations, implications, and future directions on AI in personalized learning

In addressing the research question (RQ5), which explores AI’s limitations, implications, and future directions in personalized learning, Table 6 presents a comprehensive review highlighting technical challenges and the need for further research to optimize AI usage.

Table 6 provides concrete evidence supporting the response to the research question (RQ5), detailing the limitations, implications, and future potential of AI applications in personalized learning. At the same time, Table 5 presents a clear overview of previous studies on AI-driven personalized learning. Several key challenges identified include barriers in technology integration, dependency on high-quality data, and the lack of trials in diverse educational contexts, indicating that AI-based learning systems still face significant obstacles in terms of sustainability and scalability.

Although AI is not without its shortcomings and occasionally makes errors, the implications arising from its application reveal its tremendous potential to revolutionize the educational landscape. With its ability to tailor learning content to individual needs, enhance student

engagement, and provide prompt and accurate feedback, AI demonstrates its role as a tool capable of enriching the learning experience. This technology supports a more personalized approach and offers flexible solutions to address various student ability levels, making it a crucial contributor to creating inclusive and adaptive education systems.

Further technological advancements and in-depth research on implementing AI across different educational environments are necessary to realize this potential. Future directions emphasize the importance of overcoming technical limitations, refining recommendation algorithms, and exploring emerging AI applications, such as ChatGPT, Intelligent Tutoring Systems, and more sophisticated AI-based adaptive learning systems.

4. Discussion

4.1. Challenges in AI-driven personalized learning systems

The integration of AI into personalized learning systems presents various challenges, as identified in several studies (Kanchon et al., 2024; Naseer et al., 2024; Tapalova & Zhiyenbayeva, 2022; Wang, 2023). According to research by (Huang et al., 2023), there was no significant difference in learning motivation between the control and experimental groups despite using AI-based recommendation systems. The study (Tapalova & Zhiyenbayeva, 2022) emphasizes technological infrastructure and teacher training as prerequisites for successfully adopting Artificial Intelligence in Education (AIEd). Similarly, findings (Xu et al., 2023) highlight ethical concerns related to student data collection and analysis, complicating privacy guarantees and accuracy. These challenges reflect the complexities of transitioning from traditional learning models to AI-based models, requiring attention to technological, ethical, and practical barriers.

Compared to previous research findings (Baltezarević & Baltezarević, 2024; Kanchon et al., 2024; Kucirkova & Leaton Gray, 2023; Ogata et al., 2024; Tang et al., 2020; Xiao & Yi, 2021), it is evident that while AI-based learning systems have garnered attention, their integration is still hindered by infrastructure gaps and resistance from both educators and students. The lack of adequate teacher training and robust systems to manage personalized learning content are recurring issues (Rawas, 2024; Schmid et al., 2022; Villarreal, 2023). Research by (Alsobhi & Alyoubi, 2019) also supports this, as they found limited experimental results focused solely on computer science students, without broader application across various educational contexts.

Table 6

Limitations, implications, and future Directions of AI-Driven personalized learning.

No	Author	Limitations	Implications	Future Directions
1	Huang et al. (2023)	Did not explore external factors such as learning environment support and social interaction.	AI-based recommendation systems can enhance engagement and learning motivation in flipped classrooms.	More effective systems for varying motivation levels and integration with other teaching methods are needed.
2	Tapalova and Zhiyenbayeva (2022)	The research is theoretical and descriptive, without quantitative data to assess the effectiveness of AIED.	AIED technology can adapt learning content according to individual needs, enhancing student engagement.	Further research is needed to explore the application of AIED technology in other universities through AI-based programs.
3	Tang et al. (2020)	Limitations in acquiring resources to provide more personalized learning.	Parallel innovative education can differentiate students based on ability and provide personalized guidance.	Development of adaptive learning models relevant to various contexts, including the application of AI in STEM and educational games.
4	Xiao and Yi (2021)	The integration of AI with big data and other technologies has not been fully optimized.	Personalized educational reforms can improve education quality and significantly benefit students and institutions.	Further exploration of AI applications in education, including careers and professional development.
5	Alsobhi and Alyoubi (2019)	The DAELMS adaptive system has not been tested in different educational contexts or with a more diverse user group.	Personalizing learning based on learning styles and dyslexia types can improve learning outcomes.	Further development should include more personalization variables and testing in diverse educational contexts.
6	Xu et al. (2023)	Dependence on high-quality data and challenges in integrating AI systems into existing curricula	Educators require training to use AI and ensure feedback is effective and constructive.	Further research is needed to adapt feedback systems to diverse educational contexts while addressing privacy and ethical concerns.
7	Mokmin (2020)	Limited testing on specific user groups, highlighting the need for more extensive sample trials	Potential to boost user engagement through personalized virtual trainers, aiding productivity in physical education.	Further development of accurate and personalized virtual trainers is being tested in diverse settings to ensure broad applicability.
8	Wei et al. (2021)	The assumption that student capabilities and interests remain constant over time	The application of educational psychology and AI enhances personalized learning recommendations, improving student outcomes.	Explore the impact of interactive teaching methods and focus on tailoring recommendations based on student learning experiences.
9	Bekmanova et al. (2021)	Challenges in consistently evaluating personalized learning models across contexts due to technical limitations	Personalized learning models improve teaching quality, especially for blended learning environments in higher education.	Expand models by integrating additional courses and refining assessment systems to measure material mastery more effectively.
10	Sajja et al. (2024)	Challenges in using AI teaching assistants (VTAs), including natural language understanding and LMS integration	VTAs enhance learning outcomes, student engagement, and satisfaction while fostering discussions on AI integration in education.	Develop advanced natural language processing techniques like semantic parsing and sentiment analysis to enhance adaptive learning algorithms.
11	Zhang (2022)	Personalized systems require sufficient technology infrastructure and educator training.	AI integration in education improves learning efficiency and personalizes student services, enhancing outcomes and equity.	Advance recommendation algorithms and adaptive platforms while investigating the long-term effects of these technologies in education.
12	Ogata et al. (2024)	Incomplete or non-representative data for predicting and grouping student personalities	Better personalization can improve learning effectiveness and transform educational delivery.	Further develop explainable and adaptive AI algorithms, ensuring deeper integration with educational systems to maximize accessibility and effectiveness.
13	Wang (2023)	Technical challenges integrating IoT and digital skill gaps in education.	AI recommendation models improve learning resource effectiveness, boosting student outcomes.	Research on adaptive technologies and collaboration for effective English education.
14	Ruwe and Mayweg-Paus (2023)	The rapid changes in people's attitudes and experiences toward AI systems, such as those triggered by the launch of ChatGPT, indicate that the results of this study should be considered with caution.	Trust-building is essential in feedback processes, considering the characteristics of both feedback recipients and providers.	Explore AI feedback in diverse contexts and study the impact of personalized language.
15	Kucirkova and Leaton Gray (2023)	Difficulty integrating AI with human intelligence and ensuring reliable data.	AI can enhance education, but it needs careful implementation to ensure equity.	Promote pluralistic approaches in AI development to balance power dynamics.
16	Baba et al. (2024)	Limited focus on lower Bloom's taxonomy levels and technical constraints on complex content.	AI systems like Campus + improve academic achievement, but must avoid bias.	Expand research to include higher cognitive abilities and more complex content handling.
17	Naseer et al. (2024)	Challenges in deep learning integration and technical issues in personalized pathways.	AI can boost academic performance, engagement, and satisfaction, enabling adaptive, student-centered teaching.	Research to address technical challenges and improve user experience and platform stability.
18	Kanchon et al. (2024)	The effectiveness of personalized content in improving learning outcomes needs further evaluation.	The proposed learning style detection system can integrate with standard learning systems without extra equipment.	Expand datasets and conduct A/B testing for improved model refinement and academic performance.
19	González-Rico and Lluch Sintes (2024)	Causality between variables is not fully explored, requiring deeper statistical analysis.	Technology integration with personalized guidance can improve soft skills and relevance perception, aiding future employability.	Replicate studies with larger samples and develop more adaptive pedagogical strategies.
20	Baltezarević and Baltezarević (2024)	Early stages of AI development require further resolution of issues.	AI can personalize learning and assist teachers with administrative tasks.	Explore AI's impact on creativity, interaction, and learning outcomes compared to traditional teaching.
21	Hang et al. (2024)	Focus on computer science and exclusive use of GPT-4 limits comparative insights.	AI supports personalized learning by automating MCQ creation and improving learning and performance.	Expand evaluation scope, integrate diverse subjects, and explore multiple LLMs for more complex content.
22	Guettala et al. (2024)	Ethical challenges concerning student data privacy and pedagogical implications of human elements in learning.	Generative AI can enhance personalized learning experiences, but raises urgent ethical and technical issues.	Collaboration between educators and technology revolutionizes education by embedding learning in daily life.

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Table 6 (continued)

No	Author	Limitations	Implications	Future Directions
23	Dong et al. (2024)	High reliance on quality and quantity of training data and substantial computational resources; accent and dialect generalization challenges.	Development of personalized learning tools and assessment systems for English-speaking students, utilizing generative models.	Integrate generative AI with English education, especially oral language assessment, to advance learning.
24	Annuš and Kmeť (2024)	A small sample size (N = 51) limits generalization, and no control group exists for comparison.	Personalized software enhances math knowledge, as seen in better test results compared to the control group.	Test with younger age groups to explore time use correlation and further develop software to address design and content limitations.
25	Deng et al. (2024)	Increased time costs affect feasibility and scalability in resource-limited environments.	Development of dynamic models and incremental learning algorithms to improve personalized learning paths and student experiences.	Optimize algorithms to reduce computational costs, study long-term variables, and validate model generalization across educational settings.

Addressing these challenges is crucial to making AI systems more scalable and effective in diverse settings.

Future research should explore innovative approaches to address these challenges. As suggested by (Wang, 2023), integrating the Internet of Things (IoT) and providing training for both teachers and students could offer a stronger framework for AI applications. Additionally, further investigation into the ethical implications of AI use in education and its potential biases, as proposed by (Baba et al., 2024), is necessary to ensure the fair implementation of this technology.

4.2. Opportunities in AI-driven personalized learning systems

Despite existing challenges, each finding presents substantial potential and opportunities for utilization, particularly in AI-driven personalized learning, which is vast and transformative. The study conducted by (Tapalova & Zhiyenbayeva, 2022) focuses on the potential of AI to tailor educational content to align with individual student needs, enabling learners to study according to their unique style, pace, and requirements. Similarly (Xiao & Yi, 2021), argue that AI-based systems can design personalized training plans tailored to students' career objectives. With AI-based systems, such as those used in flipped classrooms, there is an opportunity to enhance student engagement and motivation, as noted by (Huang et al., 2023). The ability to adapt the learning experience to individual preferences and abilities represents a significant opportunity to improve learning outcomes and make personalized education more interactive and meaningful.

These opportunities align with previous research findings by (Bekmanova et al., 2021; Deng et al., 2024; Kanchon et al., 2024; Ogata et al., 2024; Wang, 2023; Xu et al., 2023), which explore how personalized training models can enhance the flexibility and effectiveness of courses. AI's ability to analyze vast amounts of data and provide immediate feedback also offers unique advantages in enhancing student engagement and developing essential 21st-century skills, as shown by (Tapalova & Zhiyenbayeva, 2022). The integration of AI with educational psychology, as suggested by (Wei et al., 2021), allows for more accurate resource recommendations, helping reduce cognitive load and aligning learning materials with students' capabilities.

In order to leverage these opportunities, future research should focus on the practical application of AI skills, such as developing adaptive algorithms, recognizing student learning behavior patterns, and integrating cross-disciplinary data to create more holistic learning models (Ayanwale et al., 2024; Bauer et al., 2023). Additionally, it is crucial to explore how AI can support collaborative learning by enabling more dynamic interactions between students, teachers, and technology (Kim, 2024; Tan et al., 2022).

Future research must address ethical and privacy concerns, particularly in collecting and analyzing student data. This includes developing policies that ensure data security and algorithm transparency to ensure its use is acceptable to all parties involved. Finally, training educators to optimally leverage AI technology becomes a crucial aspect that cannot be overlooked. This is essential in equipping teachers with the necessary technical and pedagogical skills to manage AI-based learning systems.

Furthermore, research should explore the scalability of these opportunities, especially in underrepresented educational settings, and AI's role in promoting educational equity, as noted by (González-Rico & Lluch Sintes, 2024).

4.3. Benefits of AI-driven personalized learning systems

The benefits of using AI in personalized learning systems are evident in several studies showing substantial improvements in student engagement, performance, and overall learning outcomes. In particular, research by (Wei et al., 2021) found that AI-based personalized learning can significantly reduce cognitive load while improving the accuracy and adaptability of learning materials. According to (Xu et al., 2023), AI-based systems improved student performance by 95.32 %, indicating the effectiveness of personalized feedback in enhancing educational outcomes. Furthermore, studies conducted by (Alsobhi & Alyoubi, 2019) observed that AI-based personalized learning systems helped improve learning outcomes, especially for students with diverse learning styles, such as those with dyslexia.

These benefits are consistent with previous research, which shows that AI-based systems can enhance strategic thinking, problem-solving skills, and student motivation (Annuš & Kmeť, 2024; Baba et al., 2024; Huang et al., 2023; Ruwe & Mayweg-Paus, 2023; Tang et al., 2020). For instance, a study by (Huang et al., 2023) highlighted that AI integration supports a more inclusive and adaptive learning approach, facilitating the provision of materials tailored to individual preferences. This research underscores the tremendous potential of AI to accelerate educational transformation in both developed and developing countries, focusing on effectiveness, efficiency, and equity in education. Similarly (Ruwe & Mayweg-Paus, 2023), noted that students trust AI-based feedback systems more than traditional methods, contributing to enhanced trust and motivation among learners.

Research should focus on refining AI-based systems to maximize their benefits. For instance, studies could investigate integrating AI with human intelligence more effectively, balancing personalized learning with social interaction, as suggested by (Kucirkova & Leaton Gray, 2023). Additionally, exploring the long-term impacts of AI would be valuable for student retention, career readiness, and creativity, ensuring that these systems are effective in the short term and support lifelong learning and adaptability.

4.4. Limitations of AI-based personalized learning systems

The limitations of AI-based personalized learning systems are diverse, yet many stakeholders share concerns regarding the readiness and implementation of such systems across various educational sectors. For example (Huang et al., 2023), noted that AI-based personalized learning systems often overlook external factors, such as the learning environment and social interactions. These limitations may restrict the system's effectiveness outside the digital learning environment, even though this technology can potentially enhance student motivation and engagement. However, these limitations are not only technical but also

involve the preparedness of educational institutions and the skills of educators in adopting such technology.

Other research, such as that conducted by (Alsobhi & Alyoubi, 2019; Xu et al., 2023), noted that AI systems might face challenges integrating existing curricula or relying heavily on high-quality data. A lack of data or poor integration may hinder the system's ability to provide a personalized experience that effectively meets deep learning needs. Additionally, technical issues, such as an inability to address diverse learning styles or the need for better educator training, were highlighted (Mokmin, 2020; Zhang, 2022). This suggests that AI systems should be more adaptable, and educators must be sufficiently trained to use these systems effectively.

Moreover, a study by (Bekmanova et al., 2021) highlighted the difficulties in consistently evaluating personalized learning models, particularly within higher education. This raises concerns regarding the scalability and generalization of AI-based learning models, especially in higher education environments (Govea et al., 2023; Kuleto et al., 2021). The challenges of reliability and data representation for personalized learning models, as discussed by (Ogata et al., 2024), further emphasize the need for more comprehensive datasets to support adaptive AI systems.

4.5. Implications

Despite the existing limitations, AI-based personalized learning systems present significant implications for enhancing the educational experience. For example (Huang et al., 2023), found that AI-based recommendation systems could improve student engagement and motivation, particularly in flipped classroom settings, demonstrating how technology can create more flexible and adaptive learning environments, allowing students to access materials according to their needs before in-person sessions. Similarly (Tapalova & Zhiyenbayeva, 2022), emphasized that AI in education technology can adapt learning content to meet individual needs, enhancing student engagement and motivation.

Regarding broader educational reforms (Xiao & Yi, 2021), highlighted that AI's potential to personalize education could produce the best graduates from higher education, enabling them to explore professional development opportunities, including career paths in the future. This aligns with the findings of (Alsobhi & Alyoubi, 2019), who noted that personalizing learning based on learning styles or specific learning disabilities, such as dyslexia, could significantly improve learning outcomes. Furthermore, integrating AI into the educational process can help personalize learning more effectively, as AI can address various learning preferences and challenges that traditional methods may overlook.

AI also has the potential to foster a collaborative, student-centered learning environment, but to realize this potential, educator training is essential to help them understand, integrate, and effectively use this technology in the learning process. For example, research focused on physical education by (Mokmin, 2020) found that personalized virtual coaches could help increase engagement and productivity, demonstrating how AI can be applied in various educational settings. Therefore, it is crucial to develop further AI-based tools and applications that focus on developing learning content and enhancing educators' pedagogical skills in using this technology.

4.6. Future directions

Moving forward, several areas where AI-based personalized learning systems can be further developed to maximize their impact. According to (Tang et al., 2020), there is a need to develop adaptive learning models that are relevant to various educational contexts, particularly in fields such as STEM or educational games. This suggests that future research should focus on creating AI systems that are not only personalized but also contextually relevant, ensuring that the unique learning

needs of students in diverse fields such as science, technology, engineering, mathematics, and engineering (Xu & Ouyang, 2022).

Additionally, integrating AI with other technologies, such as big data and the Internet of Things (IoT), is still in its early stages and requires further optimization (J. Li et al., 2023; Paramesha et al., 2024). Research by (Xiao & Yi, 2021) proposes that future AI applications should be more comprehensive, integrating a broader range of data sources to improve the accuracy and effectiveness of personalized learning models. Research on the long-term effects of AI in education is also crucial, as highlighted by (Zhang, 2022), to understand how this technology impacts learning outcomes over time.

AI integration's ethical and technical challenges, particularly concerning student data privacy, must also be addressed (Akgun, S., & Greenhow, 2022). According to (Guetalla et al., 2024), it is crucial to balance the benefits of AI with ethical considerations when handling student data. Future research should explore methods to safeguard data privacy while still offering personalized learning experiences. Furthermore, subsequent studies are advised to focus on enhancing the adaptability of AI systems to various teaching methods, ensuring that AI enhances human intelligence rather than replacing it. This is especially important given the rapid advancement of technology, as noted by (Ruwe & Mayweg-Paus, 2023), who caution that shifts in perceptions of AI must be closely monitored.

5. Limitations of the review

This systematic review is subject to several methodological limitations. First, the search was restricted to articles indexed in the Scopus database and published between 2019 and 2024, without inclusion of other indexing services such as Web of Science, PubMed, Google Scholar, or Semantic Scholar. As such, potentially relevant studies outside the Scopus coverage particularly from across regional, non-Western, or K–12 contexts may have been inadvertently excluded. Second, only English-language peer-reviewed journal articles were included, whereas conference papers, grey literature, and articles in press were excluded to ensure publication quality. This exclusion may limit the scope of emerging or exploratory research which is often first presented in non-journal formats or regional languages.

Third, the defined five-year timeframe (2019–2024) was selected to capture recent developments in AI-enabled personalized learning; however, this may exclude foundational studies published prior to this interval. Fourth, the focus on empirical research excluded conceptual or theoretical articles, thereby omitting insights that might provide deeper understanding of ethical, pedagogical, or policy-related dimensions of AI in education.

Finally, most of the included studies reported short-term outcomes (e.g., engagement, satisfaction), with limited tracking of long-term impacts such as equity, sustainability, and cognitive development. Ethical concerns, including data privacy, algorithmic bias, and stakeholder readiness were acknowledged in several studies but rarely interrogated in depth. Future systematic reviews should expand database coverage beyond Scopus, incorporate non-English literature, and consider diverse research designs (including grey literature and preprints) to provide a more inclusive and globally representative synthesis of AI-driven personalized learning research.

6. Conclusion

This literature review uses the PRISMA framework to identify the developments and applications of AI-driven personalized learning over the past five years (2019–2024). Based on RQ1, a systematic analysis uncovered 25 key articles covering technologies such as machine learning, deep learning, natural language processing, and intelligent tutoring systems. Research distribution indicates a dominance of the Computer Science field, contributing 24 %, followed by contributions from the social sciences, engineering, medicine, and other

multidisciplinary areas.

Afterward, RQ2 explores emerging trends in AI-based learning through data visualization, with key clusters including (1) the development of AI-based tools, such as ChatGPT, to support education; (2) personalized learning using adaptive technologies; (3) concerns over ethics and privacy in AI usage; and (4) interaction between educators and AI-based systems. Based on RQ3, the findings reveal the challenges, opportunities, and benefits of AI-based learning. Despite technical and ethical barriers, applying AI under appropriate guidelines can improve educational quality, particularly in personalizing learning and developing 21st-century skills.

In response to RQ4, AI-based learning is more effective than traditional methods. These systems can enhance student performance and learning styles on an individualized basis, ensuring more profound and measurable progress. However, RQ5 identifies several limitations, such as a lack of generalization in studies, a focus on short-term outcomes, and challenges in integrating AI into traditional curricula. Nevertheless, AI has the potential to foster a more flexible and collaborative learning approach, improving student engagement and providing rapid, accurate feedback. Attention to ethical issues and educator training remains critical.

To maximize these benefits, future research should address these limitations by refining adaptive learning algorithms, expanding studies across diverse educational environments, and developing ethical policies that support AI as a tool to assist, rather than replace, human effort since AI can sometimes make mistakes, such as providing inaccurate or contextually irrelevant answers. With careful and sustained implementation, AI holds the potential to create an inclusive, adaptive, and meaningful learning environment for the future of education.

CRediT authorship contribution statement

Aprilla Fortuna: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Febri Prasetya:** Writing – original draft, Validation, Investigation, Data curation. **Agariadne Dwinggo Samala:** Writing – review & editing, Validation, Software, Resources. **Soha Rawas:** Validation, Supervision, Investigation. **Santiago Criollo-C:** Writing – review & editing, Validation, Project administration. **Deniz Kaya:** Writing – review & editing, Supervision, Project administration, Methodology. **Muhammad Raihan:** Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Data curation. **Welli Andriani:** Writing – review & editing, Writing – original draft, Resources, Data curation. **Dian Safitri:** Writing – review & editing, Validation, Investigation. **Rahmat Azis Nabawi:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

Data availability statement

This study is a systematic literature review that uses only publicly available sources; thus, no new primary data were generated or analysed. Supporting materials (screening log, PRISMA flow, extraction sheets, and analysis files) are available from the corresponding author upon reasonable request.

Declaration of the use of AI assisted technologies

The authors confirm that no generative AI or AI-assisted technologies were used to create or modify any figures or during the writing process of this manuscript. All figures were manually designed and prepared by the authors using CorelDraw 2024 and Microsoft Excel software. Grammarly, an AI-based tool, was only used for enhancing language clarity, readability, and grammar in the text, and was not involved in the creation of figures or data visualization. The authors take full responsibility for the content and accuracy of all materials presented in this manuscript.

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Ethics statement

Not applicable.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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